

AE 520 – Fall 2011 – Prof. van Leer
One-hour Exam Nr. 2 – 18 November

Only 6 days till Thanksgiving!

1. **Irreversible flow – the saga continues** (20 min, 45%) A tube filled with uniform air at rest (state 0) is closed at the left end by a piston. At time $t = 0$ the piston instantaneously accelerates to a speed $u_1 > 0$, producing a shock wave of medium strength (say, $p_1 \approx 2p_0$). After some time the piston is decelerated to a speed $u_2 \approx 0$, creating an expansion wave. See Figure 1. Once again, u_2 has been chosen such that *the final pressure p_2 is exactly equal to the original pressure p_0* .

Which statement is true,

- $$\begin{aligned} u_2 &< 0, & (1) \\ u_2 &= 0, \text{ or} & (2) \\ u_2 &> 0? & (3) \end{aligned}$$

You may have an idea what the answer is, but allow me to help you find it in a few steps.

- (a) You will need to know the relative position of the relevant Hugoniot and Poisson curves in the (u, p) -plane. On which curve will you find state (1)? [Generic answer: on the compression/expansion branch of the Poisson/Hugoniot curve through state (n) for a forward/backward-running wave.]
- (b) On which curve will you find state (2)?
- (c) Using your knowledge of the relative location of the above branches, draw a qualitatively correct (u, p) -diagram for this piston problem, with explanation.
- (d) Now answer the question regarding u_2 .

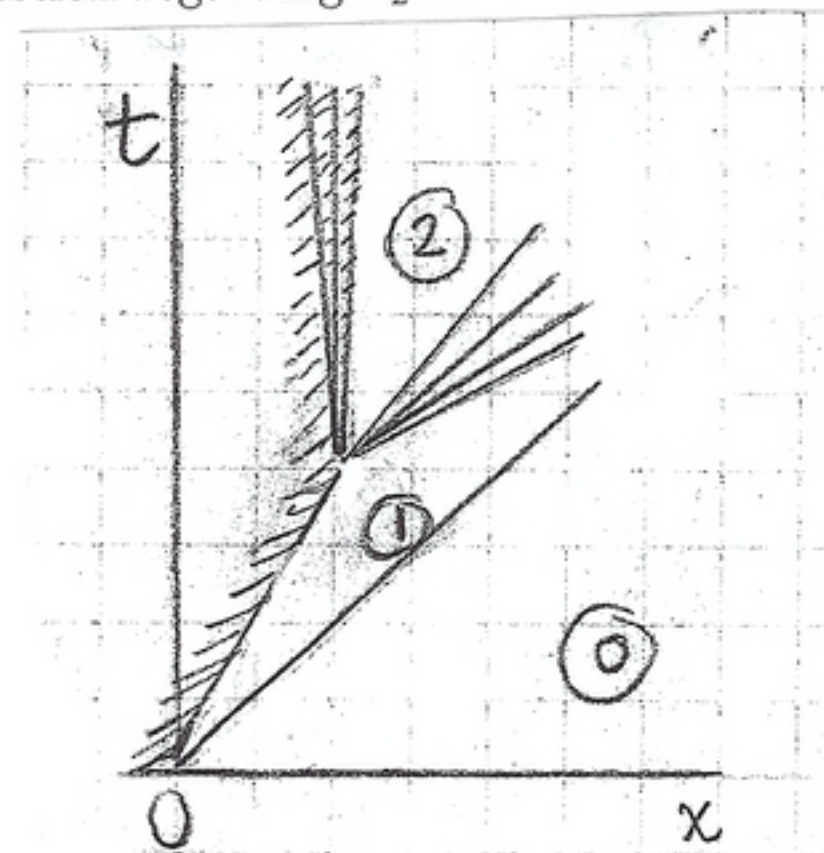


Figure 1: (x, t) -diagram.

2. A diet of Burgers (30 min, 55%)

You are going to solve the inviscid Burgers equation

$$v_t + vv_x = 0 \quad (4)$$

for the following initial-value distribution:

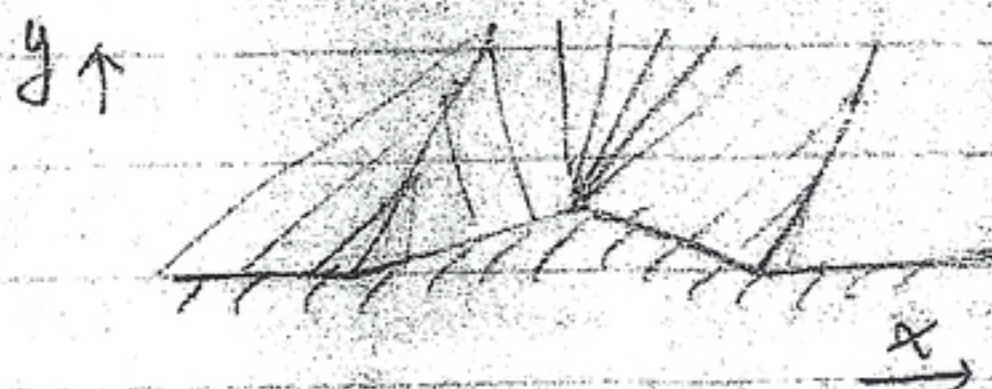
$$v(x, 0) = \begin{cases} 2 & x \leq -1, \\ 0 & -1 < x < 0, \\ 1 - x & 0 \leq x \leq 1, \\ 0 & x > 1. \end{cases} \quad (5)$$

- (a) Neatly graph the initial-value distribution. *For your own sake, don't make any mistakes here!*
- (b) Determine if the waves in this initial-value problem will interact for an infinite or finite length of time, then describe the large-time solution you expect.
- (c) Draw a neat (x, t) -diagram for the problem; take enough space to fill in details of the solution as they come along.
- (d) For some time period there are two shocks in the solution, each interacting with the expansion wave in between them. Develop an ordinary differential equation for the curved part of the left shock. Develop a similar equation for the right shock.
- (e) Solve both equations with the correct starting values.
- (f) Obtain the point where the shocks intersect. Use it to complete the (x, t) -diagram.

4.8 The "first nonlinear" description of steady 2-D supersonic flow

It is possible to derive an inviscid Burgers equation for simple waves in steady 2D supersonic flow. Let us assume the waves are created in the upper half plane by a bottom wall, as in the figure.

We assume homentropic and homenthalpic flow (s and h uniform), and, in addition,



$$K^- = \psi + \phi = K_\infty^- = \psi(M_\infty) + \phi_\infty \quad (4.8.1)$$

The remaining characteristic equation is

$$\frac{\partial(\psi - \phi)}{\partial x} + \tan(\phi + \mu) \frac{\partial(\psi - \phi)}{\partial y} = 0. \quad (4.8.2)$$

Using (4.8.1) we may rewrite this as

$$\frac{\partial(2\psi(M) - K_\infty^-)}{\partial x} + \tan(\mu(M) - \psi(M) + K_\infty^-) \frac{\partial(2\psi(M) - K_\infty^-)}{\partial y} = 0. \quad (4.8.3)$$

or

$$2 \frac{d\psi(M)}{dM} \frac{\partial M}{\partial x} + \tan(\mu(M) - \psi(M) + K_\infty^-) \cdot 2 \frac{d\psi(M)}{dM} \frac{\partial M}{\partial y} = 0 \quad (4.8.4)$$

The derivation continues in a way similar to that of Eq. (3.23) in Sec. 3.2. First, get rid of the factor $2 dv(M)/dM$ in both terms of (4.8.4):

$$\frac{\partial M}{\partial x} + \tan(\mu(M) - \nu(M) + K_{\infty}^-) \frac{\partial M}{\partial y} = 0. \quad (4.8.5)$$

The characteristic slope $\tan(\mu - \nu + K_{\infty}^-)$ is purely a function of M , let us call this function $f(M)$:

$$\tan(\mu(M) - \nu(M) + K_{\infty}^-) \equiv f(M). \quad (4.8.6)$$

Now multiply the equation by $f'(M)$:

$$f'(M) \frac{\partial M}{\partial x} + f(M) f'(M) \frac{\partial M}{\partial y} = 0, \quad (4.8.7)$$

or:

$$\frac{\partial f(M)}{\partial x} + f(M) \frac{\partial f(M)}{\partial y} = 0, \quad (4.8.8)$$

which is in the desired form. Remember that $f(M)$ is just the characteristic slope, hence, the equation for these "upward" simple waves can be written as

$$\frac{\partial \tan(\phi + \mu)}{\partial x} + \tan(\phi + \mu) \frac{\partial \tan(\phi + \mu)}{\partial y} = 0. \quad (4.8.9)$$

We may consider extending its use to situations where the underlying

assumptions are not exactly satisfied, i.e., when shocks arise. In this case both S and $V+\phi$ do not remain constant but experience jumps of the order $[p]^3$.

Neglecting these variations, we may then describe the steady shock path in (x, y) space by the jump equation associated with (4.8.9):

$$[\tan(\phi+\mu)] \left(\frac{dy}{dx} \right)_{\text{shock}} = \left[\frac{1}{2} \tan^2(\phi+\mu) \right], \quad (4.8.10)$$

or:

$$\left(\frac{dy}{dx} \right)_{\text{shock}} = \frac{1}{2} \left\{ \left(\tan(\phi+\mu) \right)_{\text{pre shock}} + \left(\tan(\phi+\mu) \right)_{\text{post shock}} \right\}. \quad (4.8.11)$$

In words: the slope of the shock path is the average of the slopes of the characteristics merging into the shock.

This opens the door to the study of wave interactions in 2D steady supersonic flow, just as in Sec. 4.5 for 1D unsteady flow. An example of an expansion-shock interaction is found in Problem Set No. 5, Problem 4.

4.9 Rotational invariance of the jump equation

Eqn. (4.8.11) has a problem: it is not rotationally invariant. If the pre- or post-shock value of $\phi + \mu$ is close to or equal to $\pi/2$, with infinite tangent, the shock angle will tend to $\pi/2$, regardless of the other value of $\phi + \mu$. This situation can be easily created by rotating the x-y frame such that the pre- or post-shock characteristics become parallel to the y-axis.

It would be highly preferable to have a formula for the shock angle that is rotationally invariant. Rather than computing the average of the tangents, it should be based on the tangent of the average:

$$\left(\frac{dy}{dx}\right)_{\text{shock}} = \tan \frac{(\phi + \mu)_{\text{pre}} + (\phi + \mu)_{\text{post}}}{2} \quad (4.9.1)$$

In order to arrive at this jump equation, we must abandon the Burgers equation (4.8.9) and derive a PDE of the form

$$\frac{\partial g(M)}{\partial x} + \frac{\partial G(M)}{\partial y} = 0 \quad (4.9.2)$$

that is equivalent to (4.8.9) in the absence of discontinuities. The jump equation associated

with (4.9.2) is

$$\left(\frac{dy}{dx}\right)_{dh} = \frac{\Delta G}{\Delta g}, \quad (4.9.3)$$

which will not yield (4.8.11) unless $g(M) = f(M)$, $G(M) = \frac{1}{2}(f(M))^2$. For (4.9.2) and (4.8.8) or (4.8.9) to be equivalent it is sufficient to require that

$$G'(M) = g'(M) f(M), \quad (4.9.4)$$

as can be seen by multiplying (4.8.5) with $g'(M)$.

The notion that equivalent PDE's lead to different jump equations may be a surprise, but has been encountered earlier. Recall that we derived a conservation law for the entropy (Sec. 1.6, Eqn. (1.6.1)) by combining the entropy-advection equation

$$S_t + u S_x = 0, \quad (4.9.5)$$

valid for adiabatic reversible flow, with the mass-conservation law

$$\rho_t + (\rho u)_x = 0, \quad (4.9.6)$$

i.e., $\rho \times \text{Eqn. (4.9.5)} + S \times \text{Eqn. (4.9.6)}$ gives

$$(\rho S)_t + (\rho S u)_x = 0 \quad (4.9.7) \\ = (1.6.1)$$

If, more generally, we multiply (4.9.5) by $\rho \sigma'(S)$, where $\sigma(S)$ is a well-behaved function of the entropy, and Eqn. (4.9.6) by $\sigma'(S)$, then adding the two yields

$$(\rho \sigma(S))_t + (\rho \sigma(S) u)_x = 0, \quad (4.9.8)$$

a conservation law associated with a jump equation different from (4.9.7). So here we have an infinity of jump equations, all derived from the same PDE, and all wrong, because Eqn. (4.9.5), expressing conservation of entropy along a particle path, is not true across a shock. If the pre-shock state and the shock speed are given, the post-shock state must be determined from the jump equations for density, momentum density and energy density. The post-shock entropy will then follow as a dependent variable; inserting it into the jump equation associated with (4.9.8) will not return the correct shock speed.

Returning to 2D steady flow in the simple-wave approximation: none of the jump equations of the form (4.9.3) is exactly valid, because the entropy and one Riemann invariant have been given a fixed value.

But they are of acceptable accuracy as long as the shocks described are sufficiently weak. Among these reasonably accurate formulas we would still do best by selecting the one that is rotationally invariant. (NB: for 1-D time-dependent flow the equivalent choice is a Galilei-invariant formula, which we found without effort;

$$\left(\frac{dx}{dt}\right)_{sh} = \frac{(u+a)_{pre} + (u+a)_{post}}{2}; \quad (4.9.9)$$

(cf. Eqs. (4.1.4) and (4.1.5).)

Working backwards from Eqs. (4.9.3) and (4.9.4) toward (4.9.1) we find the PDE that satisfies our constraints:

$$\frac{\partial}{\partial x} \sin(\phi + \mu) - \frac{\partial}{\partial y} \cos(\phi + \mu) = 0. \quad (4.9.10)$$

That this is equivalent to (4.8.5), hence to (4.8.8), follows from differentiating (4.9.10) with respect to $\phi + \mu$ and then with respect to M :

$$\cos(\phi + \mu) \frac{d(\phi + \mu)}{dM} \frac{\partial M}{\partial x} + \sin(\phi + \mu) \frac{d(\phi + \mu)}{dM} \frac{\partial M}{\partial y} = 0; \quad (4.9.11)$$

here $\phi + \mu$ is to be regarded as a function of M :

$$\phi + \mu(M) = \phi + \gamma(M) + \mu(M) - \chi(M) = K + \mu(M) - \chi(M). \quad (4.9.12)$$

The jump equation reduces to the desired result:

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$$\left(\frac{dy}{dx}\right)_{sh} = \frac{-\Delta \cos(\phi + \mu)}{\Delta \sin(\phi + \mu)} = \frac{2 \sin \langle \phi + \mu \rangle \sin \frac{\Delta(\phi + \mu)}{2}}{2 \cos \langle \phi + \mu \rangle \sin \frac{\Delta(\phi + \mu)}{2}} \\ = \tan \langle \phi + \mu \rangle.$$

The rotational invariance of this formula suggests that it is most practical to use it in polar coordinates rather than Cartesian coordinates — you'll find out quickly when exploring the latter choice. That's why in Problem 5.6 you are instructed to derive the shock path in the shock-expansion interaction in polar coordinates (r, θ) , with the origin in the center of the expansion wave. Thus, along each characteristic Γ^+ of the expansion fan we have

$$\vartheta = \phi + \mu.$$

The shock always bisects the angle between the Γ^+ characteristics that merge into the shock; calling the angle between shock and x-axis σ we have

$$\sigma = \langle \phi + \mu \rangle.$$

To find an ODE for the shock path, take an element AB of the shock path (see illustration) and consider triangle ABC .

$$\angle ABC = m_3 - \sigma = \sigma - \vartheta = \frac{m_3 - \vartheta}{2}$$

This information should suffice for you to finish problem 5.6.
